

Master of Sciences AI DSM

Department of Digital Management

MSc Artificial Intelligence and Digital Strategy Management

Syllabi 2026-2027

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FOREWORD

Educating the Leaders of the AI Era

Artificial Intelligence is not simply another technological innovation. It represents a profound shift in the way organizations create value, make decisions, compete, and innovate. As previous industrial revolutions transformed production and globalization reshaped markets, AI is now redefining the foundations of management itself.
The MSc in Artificial Intelligence and Digital Strategy Management was designed around one fundamental conviction:

Tomorrow's competitive advantage will belong to leaders who understand technological change; not simply those who master today's tools.

Programming languages, software platforms and AI applications evolve at an unprecedented pace. A technology considered state-of-the-art today may become obsolete within months.

For this reason, this MSc does not aim to train software engineers or AI developers.
Instead, it prepares students to become strategic managers, decision-makers, innovators and transformation leaders capable of navigating continuous technological disruption.

Rather than developing expertise in a single technology, students build a deep understanding of the principles that drive digital transformation, enabling them to anticipate emerging trends and make informed strategic decisions in uncertain environments

Our Vision

The program prepares future leaders who are able to:

- Understand the strategic impact of Artificial Intelligence on organizations and industries.
- Anticipate technological evolution rather than react to it.
- Transform data into strategic business decisions.
- Lead multidisciplinary teams through digital transformation.
- Create sustainable competitive advantage through innovation.

From Data to Strategic Decision-Making

Traditional management has long relied on analysing historical performance to guide future decisions. *Artificial Intelligence fundamentally changes this paradigm.*

Managers are now expected to:
- interpret complex data ecosystems;
- leverage predictive analytics;
- identify weak signals and emerging trends;
- simulate future scenarios;
- make proactive rather than reactive decisions.

The objective is therefore not only to understand data, but to develop a forward-looking strategic mindset where decisions are increasingly supported by AI-driven insights and predictive models.

Technology as a Means, Leadership as the Goal

One of the distinctive characteristics of this MSc is its balanced approach between business strategy and technological literacy.

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Students are not expected to become developers.
Instead, they develop sufficient technical understanding to:

- communicate effectively with software engineers and data scientists;
- understand the architecture of AI systems;
- evaluate technological solutions;
- estimate project complexity and implementation costs;
- manage digital projects efficiently;
- bridge the gap between business needs and technical teams.

This technological culture enables future managers to make informed decisions while allowing technical experts to focus on implementation.

Leading People in the Age of AI

Digital transformation is ultimately a human transformation.
Tomorrow's organizations will increasingly rely on teams composed of professionals with highly diverse backgrounds, cultures and expertise.

Successful managers must therefore be capable of creating collaborative environments where technical specialists, business experts and creative profiles work together effectively.

For this reason, the program places particular emphasis on:

- inclusive leadership;
- intercultural management;
- change management;
- collaborative decision-making;
- ethical and responsible innovation.

The objective is to develop leaders capable of attracting diverse talents, building high-performing teams and guiding organizations through continuous transformation.

Learning by Doing

Academic knowledge is continuously connected with professional practice.

Throughout the programme, students experience learning through:

- real-world business cases;
- AI and digital transformation projects;
- hands-on workshops;
- company challenges;
- Design Lab experimentation;
- conferences and industry events;
- guest lectures delivered by AI practitioners and executives;
- an international study trip within a leading digital innovation ecosystem.

This experiential approach ensures that graduates can transform concepts into actionable business strategies.

PROGRAM OBJECTIVES

Upon completion of the MSc, graduates will be able to:

Strategic Thinking

- Design AI-enabled business strategies.
- Analyse digital business models.
- Support organizational transformation.

Data & AI Literacy

- Understand the foundations of Artificial Intelligence.
- Interpret data using business analytics.
- Leverage predictive approaches for decision-making.

Innovation

- Develop innovative digital products and services.
- Apply design thinking methodologies.
- Foster entrepreneurial thinking.

Leadership

- Lead multidisciplinary and multicultural teams.
- Manage AI-driven transformation projects.
- Support organizational change.

Ethics & Governance

- Assess ethical implications of AI.
- Integrate responsible innovation into business strategy.
- Understand governance challenges related to emerging technologies.

WHO TEACHS THIS PROGRAMME?

The MSc brings together a diverse faculty composed of:

Academic Faculty

Professors from BSB and partner universities providing rigorous theoretical foundations and research-based insights.

Industry Experts

Executives, consultants, entrepreneurs and AI specialists sharing practical experience from real transformation projects.

Guest Speakers

International professionals bringing current perspectives on innovation, digital strategy and emerging technologies.

This combination ensures that students benefit from both academic excellence and practical relevance.

OUR TEACHING PHILOSOPHY

The MSc adopts a practice-oriented, project-based and experiential learning approach, combining academic rigor with professional application.

Students learn through:

- ✓ Business case studies
- ✓ Hands-on AI workshops
- ✓ Python and data analytics laboratories

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- ✓ Generative AI experimentation
- ✓ No-code development
- ✓ Design Thinking sessions
- ✓ Company consulting projects
- ✓ Collaborative teamwork
- ✓ International exposure
- ✓ Professional conferences
- ✓ Guest lectures from industry leaders

Learning is designed to encourage curiosity, experimentation, critical thinking and continuous adaptation; essential qualities for leading organizations in an AI-driven world.

Academic Excellence with a Global Vision

The MSc in Artificial Intelligence and Digital Strategy Management reflects the academic ambition and international excellence of Burgundy School of Business (BSB), one of the select groups worldwide to hold the prestigious "Triple Crown" accreditation: EQUIS, AACSB, and AMBA. This distinction, achieved by fewer than 1% of business schools globally, recognizes the highest standards of academic quality, faculty excellence, internationalization, and continuous innovation in management education.

BSB was the first business school in the Lyon area to launch a master's program placing Artificial Intelligence at the heart of management education. From its inception, the program was designed with a clear vision: to prepare future leaders capable of understanding and shaping the profound transformation brought about by AI across organizations, industries, and society.

As the AI landscape has evolved at an unprecedented pace, this academic track has undergone a comprehensive redesign to integrate the latest technological and strategic developments. Beyond the foundations of machine learning and data-driven decision-making, the curriculum now incorporates the new paradigms transforming business and management, including Large Language Models (LLMs), autonomous AI systems, intelligent agents, generative AI ecosystems, and emerging human-AI collaboration models. This continuous evolution ensures that graduates are prepared not only for today's digital economy but also for the innovations that will define the organizations of tomorrow.

Our expertise has established BSB as a recognized academic actor in AI-driven management education. More than a curriculum, it represents an innovative educational model combining academic rigor, practical application, interdisciplinary learning, and responsible leadership. This pedagogical approach has proven both robust and adaptable, making it a model that can be shared and deployed internationally while remaining fully aligned with the highest standards of higher education.

PEDAGOCIAL TRACK

COURSES – FALL SEMESTER	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER
AI in organization				
Strategic Foresight for Business Management				
Data Sciences applied to Organization Decision				
Big Data & Customer Insights				
Data Visualization and Story Telling				
Advanced Prompting and API integration				
Research Methodology				

COURSES – SPRING SEMESTER	JANUARY	FEBRUARY	MARCH	APRIL
Anticipatory Governance				
AI in Business				
Growth Marketing				
Supply Chain				
Generative AI Management				
Autonomous Systems				
Automation with Low Code				
Agile Project Management				
Field Trip				
Research Methdology				

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Advanced Prompting & API Integration

MGE AC 05 03 Elaborer une vision stratégique responsable, fixer les objectifs en intégrant les enjeux managériaux et de gouvernance et proposer des outils de pilotage

Semester	Contact Hours	ECTS Credits	Programme Competency
Semester 1 (Fall)	18 hours (6 sessions x 3h)	3 ECTS	MGE AC 05 03

1. Course Overview

Prerequisites

Basic digital literacy. Python eLearning class on Data Camp is recommended as a co-requisite to support light scripting during API workshops but is not a strict requirement (optional).

Course Description

This course develops the technical-managerial skill of designing and optimizing AI-powered workflows through advanced prompt engineering and API integration.

Students learn to move beyond ad hoc prompting toward structured, reusable prompting methods, and to connect generative AI capabilities into broader digital workflows via application programming interfaces.

The course covers prompt design principles such as role, context, constraints, format specification and chain-of-thought structuring, prompt evaluation and iteration methods, and the fundamentals of API-based integration, including authentication, request/response structures and chaining multiple services. Students apply these skills to build a small, automated workflow that solves a real business problem, then write an individual synthesis assessing the workflow's business impact and proposing corrective actions for cost, quality or reliability issues.

The course is intentionally positioned as the technical-managerial bridge between strategic AI courses and hands-on technical work: it does not cover the strategic question of which generative AI use cases to prioritize, addressed in Generative AI Management, nor agentic or autonomous system design, addressed in Autonomous Systems, but focuses on the craft of building reliable, optimized AI-to-API workflows.

2. Teaching and Learning Approach

Pedagogical Approach

The course follows a hands-on workshop format in which each session combines a short technique briefing with an immediate applied exercise. Students build incrementally toward a working workflow, progressing from single prompt design to prompt chains to simple API integration. Peer workflow review sessions surface optimization opportunities, and the course favors live demonstration and iterative debugging over lecture-style teaching.

- **Applied exercises**: prompt drafting and iteration
- **Workshops**: API integration build
- **Peer review:** structured workflow review sessions
- **Individual project**: final workflow and synthesis note

Teaching Resources

- Generative AI tools with API access
- Lightweight workflow/automation environments supporting API calls
- Prompt-engineering pattern libraries
- Sample API documentation (public, non-sensitive APIs)
- Cost and quality tracking templates

3. Assessment

Program competency MGE AC 05 03: Develop a responsible strategic vision, set objectives that integrate managerial and governance challenges, and propose steering tools.

Assessment Strategy [60% individual + 40% collective]

Assessment centers on an individual technical project to ensure each student demonstrates personal mastery of prompting and API integration, complemented by a synthesis note assessing business impact, consistent with the course's bridge role between strategy and technique.

ASSESSMENT	TYPE	WEIGHT
Individual synthesis note - business impact and corrective actions	Individual	60%
Individual technical project - AI workflow with API integration	Collective	40%

BELOW EXPECTATIONS	MEETS EXPECTATIONS	ABOVE EXPECTATIONS
Ineffective prompts and limited or incomplete technical integration, with little or no impact on the workflow; a synthesis note lacking critical distance.	Structured prompts and simple API integration meeting the functional need, with a meaningful workflow impact, supported by argumentation in the synthesis note.	A high-performing automated workflow integrating multiple APIs, with cost and quality optimization supported by measured indicators; a synthesis note integrating steering and concrete, relevant corrective actions.

4. Detailed Course Schedule

SESSION 1 Beyond Ad Hoc Prompting: Principles of Structured Prompt Design	
Learning Outcomes	Compare alternative prompting and integration strategies for a given functional requirement.
Competencies developed	Apply; Analyze
Content	Anatomy of an effective prompt, including role, context, constraints and output format; common failure modes; introduction to the workflow project brief.
Learning activities	Applied exercise: rewrite a weak prompt into a structured one and compare outputs.
Deliverable / preparation for next session	Select the business task for the individual workflow project.
SESSION 2 **Prompt Chains and Iterative Refinement**	
Learning Outcomes	Apply structured prompt-design principles to produce reliable, reusable AI outputs
Competencies developed	Apply; Develop; Evaluate
Content	Multi-step prompt chains; few-shot examples; iterative testing and refinement methodology; defining quality criteria.
Learning activities	Workshop: build a multi-step prompt chain for the selected business task.
Deliverable	Submit a draft prompt chain with documented test results.
SESSION 3 **API Fundamentals for Non-Engineers**	
Learning Outcomes	Design a multi-step AI workflow integrating at least one external API.
Competencies developed	Apply; Analyze
Content	What an API is; requests, responses and authentication basics; reading API documentation; connecting a generative AI tool to an external service via API.
Learning activities	Guided workshop performing a first API call that integrates a generative AI output with an external service.
Deliverable / preparation for next session	Produce a working minimal API integration.

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SESSION 4 Designing a Multi-Step AI-to-API Workflow	
Learning Outcomes	Develop an optimized, documented AI-to-API workflow addressing a real business task.
Competencies developed	Design; Apply; Develop
Content	Chaining multiple API calls; basic error handling; designing for reliability.
Learning activities	Workshop: extend the project into a multi-step workflow combining prompting and API calls.
Deliverable	Submit a draft end-to-end workflow.
SESSION 5 **Optimizing Cost, Quality and Reliability**	
Learning Outcomes	Evaluate prompt and workflow performance using defined quality and cost indicators. 5. Analyze the trade-offs between prompting sophistication, cost and output quality.
Competencies developed	Evaluate; Compare; Critically assess
Content	Cost drivers in AI-to-API workflows; trade-offs between model/prompt sophistication and cost; quality monitoring approaches.
Learning activities	Peer workflow review session; optimization exercise applied to the project workflow.
Deliverable / preparation for next session	Submit an optimized workflow version with documented metrics.
SESSION 6 **Final Workflow Demonstrations and Business Impact Synthesis**	
Learning Outcomes	Critically assess the business impact and risks of an AI-powered workflow and propose corrective actions.
Competencies developed	Develop; Critically assess; Analyze
Content	Presenting technical workflows to a business audience; structuring the individual synthesis note on impact and corrective actions.
Learning activities	Individual workflow demonstrations with instructor and peer feedback.
Deliverable	Submit the final workflow and individual synthesis note.

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5. Overlap Prevention

EXCLUDED TOPIC	COVERED IN
Generative AI use-case prioritization and organizational strategy	Generative AI Management
Agentic AI and multi-agent architecture design No-code/low-code business process automation platforms AI governance and regulatory compliance	Autonomous Systems (LLM's, Agentic AI, Robotics) Automation with low code Anticipatory governance: policy, ethics and adaptive systems

6. Core References

Essential References

- White, J., et al. (2023). A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. arXiv preprint.
- Liu, P., et al. (2023). Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in NLP. ACM Computing Surveys.
- Ekin, S. (2023). Prompt Engineering for ChatGPT: A Quick Guide to Techniques, Tips, and Best Practices. TechRxiv.
- Richardson, L., & Amundsen, M. (2013). RESTful Web APIs (selected chapters). O'Reilly Media.
- Mollick, E. (2023-2024). One Useful Thing (newsletter), selected practitioner essays on prompting and workflows.

Additional Resources

- Official prompting documentation from major generative AI providers
- Postman API learning resources
- Zapier/Make workflow-automation learning guides (conceptual reference)

Master of Sciences
AI and Digital Strategy Management

DIGITAL MANAGEMENT DEPARTMENT

CAMPUS DE LYON, FRANCE

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